

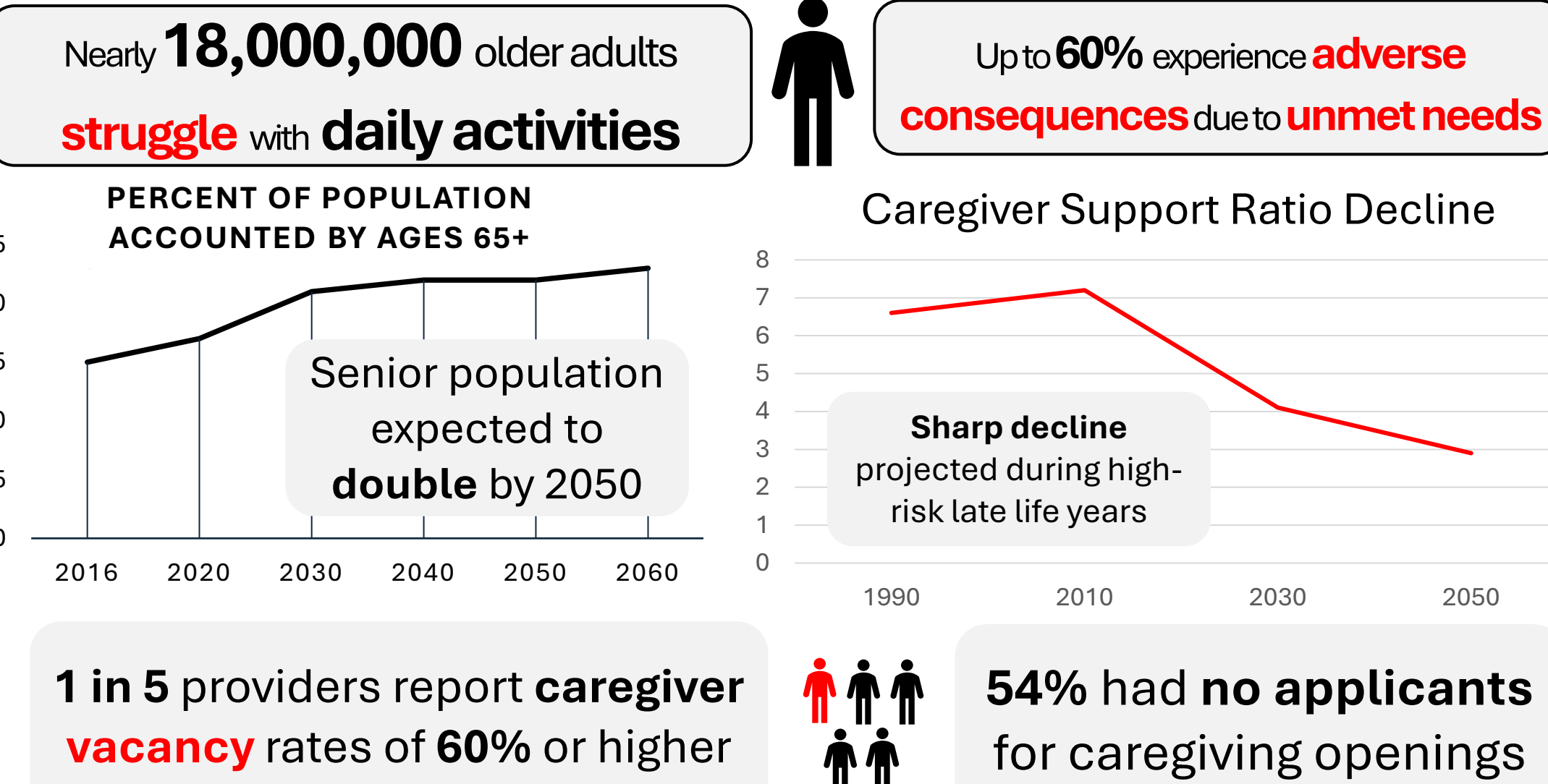
An Intelligent Bimanual Mobile Manipulator for Assisted Living Applications

Introduction

Problem Statement

With the growing national **shortage** of caregivers, there is a need for solutions to **help individuals with common household tasks**. We aim to develop a low-cost, **mobile robotic manipulator system** to enhance quality of life, independence, and accessibility for those in need of support.

Background Information



Existing Solutions / Related Work

Stretch	TIAGo	Fetch	Mobile ALOHA	TidyBot++
✗ Dextrous	✗ Dextrous	✗ Dextrous	✓ Dextrous	✓ Dextrous
✓ Mobile	✓ Mobile	✓ Mobile	✓ Mobile	✓ Mobile
✓ Accessible	✗ Accessible	✗ Accessible	✗ Accessible	✗ Accessible

Engineering Objectives

Research Goal

Develop a mobile manipulator robot system that possesses sufficient **dexterity, mobility, and usability** to effectively aid users in object retrieval and delivery tasks.

Criteria

Mobility

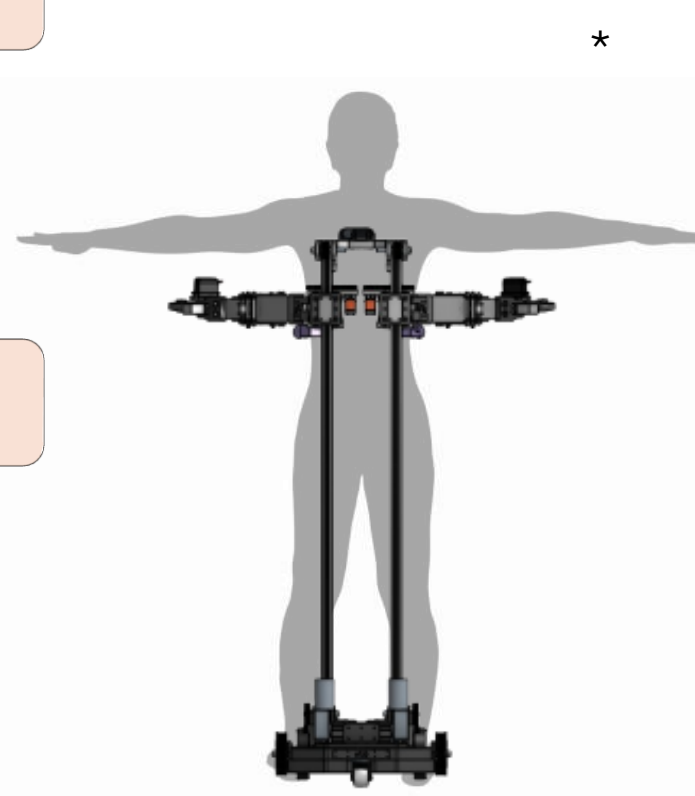
- Base dimensions less than 350mm x 350mm
- Arm max extension length at least 600mm
- Lift mechanism max height at least 800mm

Dexterity

- Shoulder joint range of motion greater than 45°
- Utilizes a 2 degree of freedom differential wrist
- Manipulation of objects greater than Ø70mm
- Manipulation of objects greater than 200g in mass

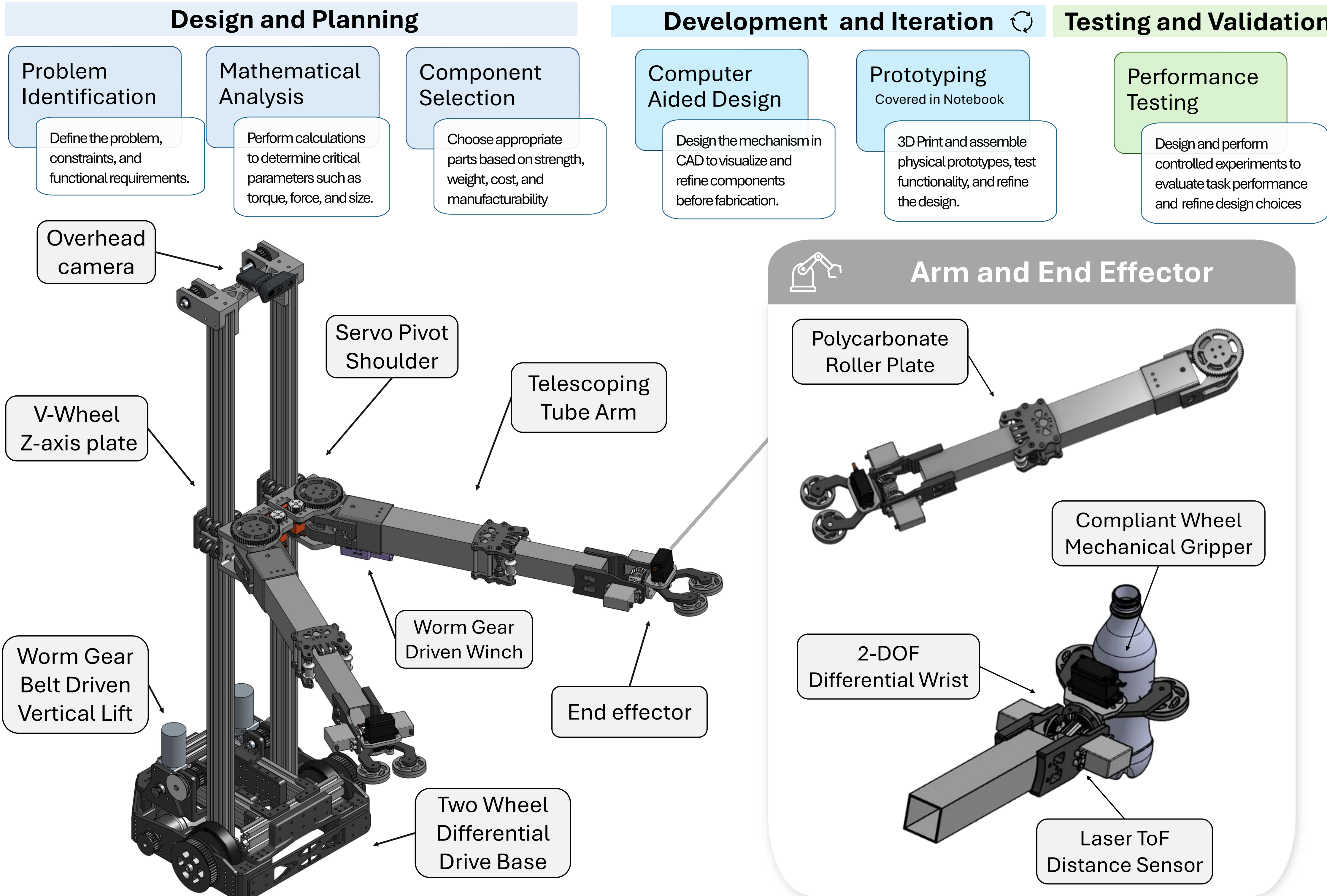
Usability

- Capable of remote teleoperation
- Joint control interface must be simple but maintain smooth, precise motion control
- Utilizes computer vision and YOLOv5
- Utilizes wheel odometry and time-of-flight (ToF) sensors for autonomous aid
- Sufficient battery life for 5 tasks before charging

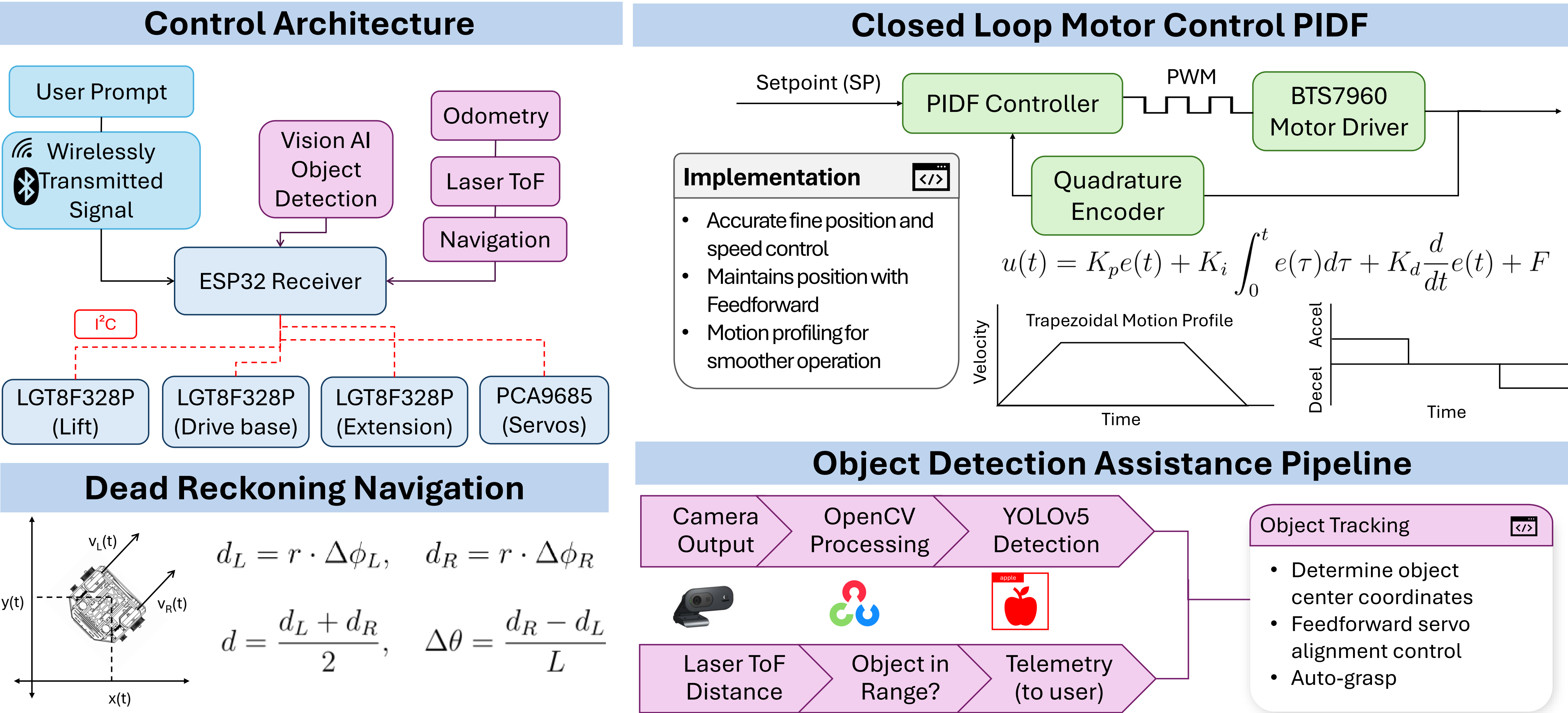


Systems Overview

Engineering Design Process



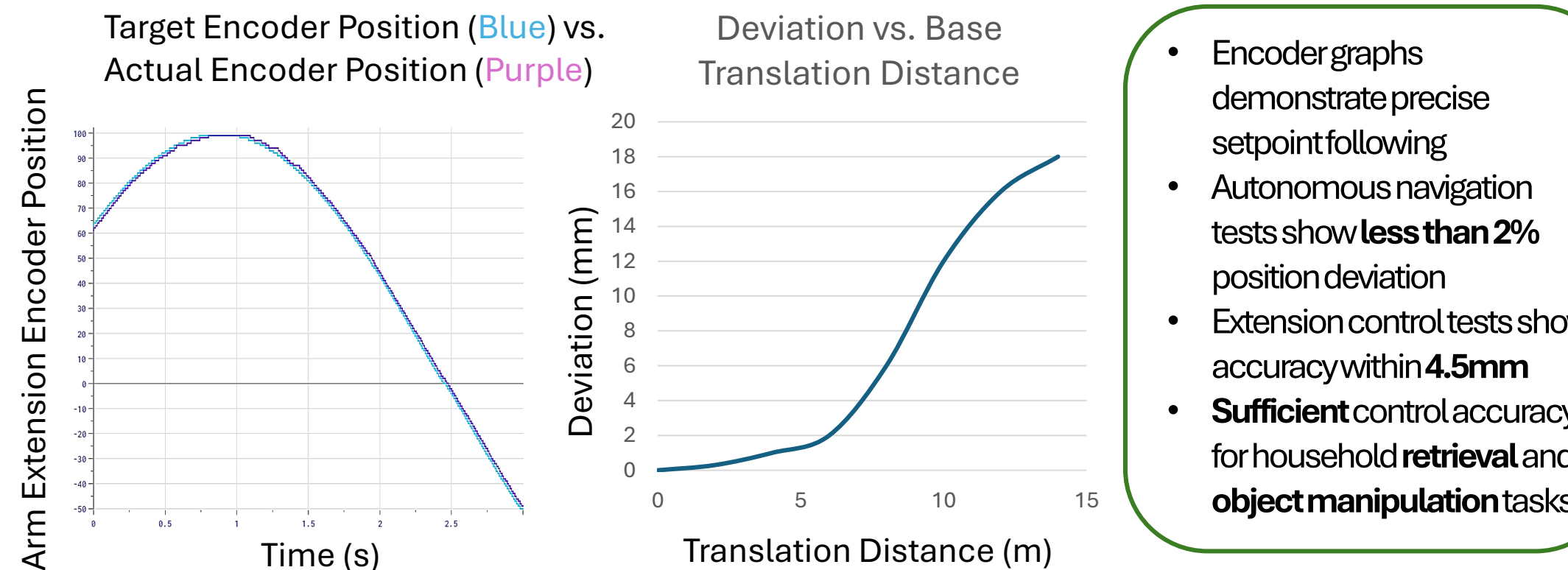
Software and Control



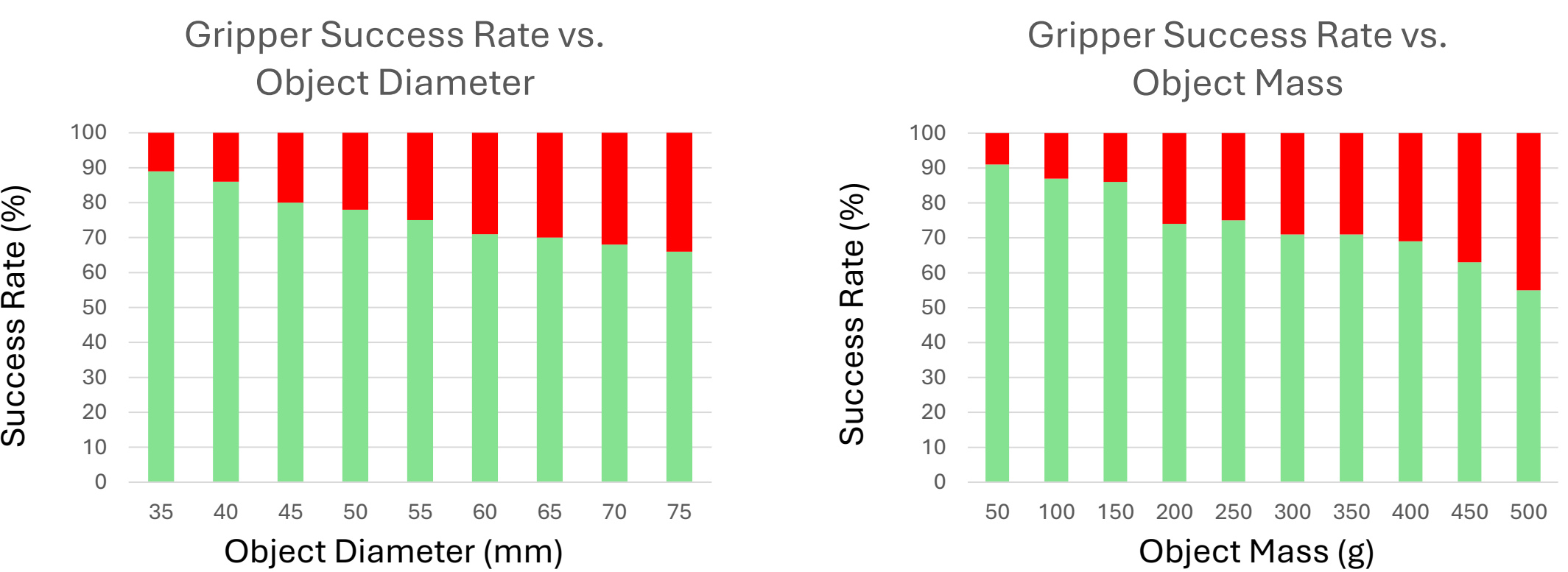
Results

Below are results from the testing of individual subsystems' performance; experiments were designed with the goals of determining **real-world application feasibility**, as well as **benchmarking** compared to the defined **engineering criteria**.

Control Accuracy



Mechanism Capabilities



Conclusions

We present a **novel bimanual mobile manipulator** designed for assisted living, integrating an **intuitive AI-assisted teleoperation** protocol and **autonomous control**. Testing showed effective mobility and dexterity, demonstrating reliable movement, object manipulation, and control accuracy.

Future Work

- Implementation** of a LLM (Large-Language Model) and Imitation Learning for enhanced user-specific accessibility
- Implementation** of LiDAR simultaneous localization and mapping (SLAM) to improve autonomous pathfinding/planning.

Key References

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Full Bibliography available in Research Notebook

Unless stated otherwise, all images were generated by the student researcher.

The student researcher would like to acknowledge Dr. Kimberly Cho (The Pembroke Hill School) and Ms. Mina Bradley for their invaluable support. He would also like to thank Mr. Hongbiao Zhao and Mrs. Minfang Wu for always pushing him to his best and putting up with his late-night procrastination.